



Ethics of Synthetic Biology



Foreword

1. Background and Objectives

With the rapid development of synthetic biology, it has demonstrated immense potential in addressing global environmental challenges, particularly in water pollution and marine governance. From engineered bacteria that efficiently degrade industrial wastewater, to modified organisms for marine environmental remediation, and innovative strategies for precisely monitoring water pollution and preventing harmful algal blooms like red tides, synthetic biology offers us unprecedented tools to protect our blue planet.

However, as this emerging technology moves toward practical application, it is also accompanied by profound ethical, social, and safety challenges. We must carefully consider: What ecological risks might arise from releasing engineered organisms into open water environments? How can we balance technological innovation with potentially irreversible impacts? How should the benefits and risks brought by the technology be distributed equitably among different social groups? These questions are increasingly drawing widespread attention from the international community.

The International Genetically Engineered Machine (iGEM) competition, as a premier international science and technology contest in the field of synthetic biology, brings together elite teams from numerous universities and research institutions and has always encouraged participants to integrate social responsibility into their scientific practices. It is based on this consensus that several iGEM teams, including SCU-China, XJTLU-China, ZJU-China, and SUSTech-OCEAN, have decided to collaborate on writing this book, *The Ethics of Synthetic Biology*.

This book focuses on the application of synthetic biology in water environment governance, with the content of each chapter originating from the practical work and in-depth reflections of our respective teams' projects this year. By exploring the ethical dilemmas in specific scenarios such as wastewater treatment, marine ecological intervention, pollution monitoring, and red tide prevention, we aim to provide an ethical framework focused on environmental applications for researchers, policymakers, ethics review boards, and the public in China and around the world. Our goal is to jointly promote the development of synthetic biology in a more responsible, equitable, and sustainable direction.

2. Editorial Team

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Environmental Protection and Synthetic Biology Initiative

—— Perspective of Wastewater Treatment

Written by SCU-China

1. Analysis of application efficiency and adaptation scenarios of synthetic biological sewage treatment technology

With the accelerated development of urbanization and industrialization, sewage treatment is no longer a simple technical problem, but a systemic challenge related to public health, ecological security, and resource circulation. Eutrophication, red tide, and hypoxic events caused by excessive nutrients are becoming more frequent, and trace organic pollutants (including drug residues, endocrine disruptors, and aromatic hydrocarbons) are deposited in the environment for a long time, which is difficult to completely remove by conventional treatment processes. At the same time, traditional treatment processes incur a significant cost in terms of energy consumption, chemical usage, and sludge disposal. In the face of limited energy and growing water resource pressure, the field of sewage treatment urgently needs to find a new balance between "efficient decontamination, low-carbon energy consumption, resource recovery, and controllable safety", which provides an important opportunity for the application of synthetic biology technology.

1.1 The application status and structural limitations of traditional sewage treatment technology

After decades of development in the field of sewage treatment, traditional chemical and engineering methods have formed a relatively mature technical system, which still occupies a dominant position in the short-term removal of suspended solids and general organic loads. The chemical precipitation method is done by adding metal salts such as aluminum salts $\text{Al}_2(\text{SO}_4)_3$, PAC, or iron salts $\text{FeCl}_3/\text{FeSO}_4$, alkali or sulfide, so that soluble nutrients or metals form insoluble colloids/solid-liquid separation after precipitation, including successful cases such as the use of aluminum salts to control endogenous phosphorus in lakes in Sweden and North America, its

process is mature and simple, and it is very effective for high-concentration pollutants and short-term fast-acting phosphorus removal. The ion exchange method uses functionalized resins (cation/anionic resins, chelated resins, etc.) to selectively enrich and adsorb the target ions. The resin can be chemically regenerated, has high selectivity, and can efficiently remove the target ions at low concentrations, with almost no volumetric sludge being produced.

As a modern treatment method, membrane separation technology separates particles and target solutes under pressure or an electric field driven by the pore size/selectivity of the membrane through different types of membrane materials such as MF/UF/NF/RO/ED/MD, with high separation accuracy, stable water quality, and easy automation and modular design. Typical success stories include the Tianjin desalination project in China and the UF→RO reuse system of a textile enterprise in Zhejiang/Jiangsu (multiple demonstrations from 2015 to 2023, reuse rate of 50-60%+), and the large-scale seawater RO plant in Kwinana, Australia (commissioned in 2006-2007, with a capacity of 140,000 m³/d, including energy recovery units), which demonstrate the feasibility of membrane technology in large-scale applications.

However, traditional solutions face insurmountable structural shortcomings, limiting their further development in future wastewater treatment. Chemical precipitation will produce a large amount of sludge that needs to be dehydrated and disposed of, which is prone to clogging and corrosion, and the sedimentation stability is greatly affected by pH and redox conditions. The cost of ultra-deep treatment of low-concentration pollutants is high. Ion exchange resins face the problem of high equipment capital and regeneration costs, and are sensitive to suspended solids, organic matter, and oils (easy to clog/passivate). The regeneration process produces high concentrations of regeneration waste liquid that needs to be further treated. Although membrane separation technology is exact, it faces core challenges, including membrane contamination/scaling/biofilm, as well as low throughput and lifetime, high membrane element and maintenance costs. Furthermore, treating highly suspended or high-salt wastewater requires fine pretreatment and additional energy consumption. In addition, advanced oxidation is often accompanied by high energy consumption and complex operating conditions, which may produce intermediate products, while adsorption and regeneration bring secondary disposal problems, insufficient selectivity for low-concentration complex mixed pollutants, high long-term economic cost, and limited resource utilization potential.

1.2 Innovative breakthroughs and multi-scenario applications of synthetic biological sewage treatment technology

Faced with the limitations of traditional technologies, synthetic biology offers a series of innovative solutions for wastewater treatment, which demonstrate significant performance advantages and adaptability in multiple application scenarios. The ecologically engineered Living Machine is especially suitable for low-energy treatment needs at the community, campus, and scenic spot levels through multi-level biological processes such as plants, microorganisms, and sediment. In several demonstration projects, inlet and outlet data show that BOD and TSS removal rates are usually 90-99%, nitrate and organic nitrogen are significantly reduced, and the system has both landscape and educational functions, providing an eco-friendly solution for distributed wastewater treatment.

For refractory organic pollutants in industrial wastewater, the targeted degradation route of engineered bacteria demonstrates precise and efficient treatment abilities. Engineered microorganisms, such as the *Pseudomonas* series or metabolically modified strains, can remove benzene (BTX) and refractory aromatics from industrial wastewater. In immobilized reactors, the short-residence time removal rate of toluene compounds is typically 60-80%, and the mineralization rate can reach nearly 95% after process optimization or extended contact time. Through chromosomal integration and metabolic homeostasis design, these engineered bacteria can enhance genetic stability, and studies have shown that they can sustain functionality over long-term passage, offering technical support for continuous and stable industrial wastewater treatment.

In the treatment and recycling of urban domestic sewage and animal husbandry tailwater, the microalgae-driven HRAP (high-throughput algae pond) technology has outstanding performance. $\text{NH}_4\text{-N}$ removal rates are often between 80-100%, PO_4^{3-} removal fluctuates between 30-80%, COD removal is about 40-80%, and biomass can be produced at the level of about $10\text{-}20\text{ g VSS m}^{-2}\text{ d}^{-1}$ for fertilizer or energy recovery. This technology not only realizes sewage purification but also realizes resource recycling through biomass recycling, reflecting the concept of the circular economy.

For trace amounts of difficult-to-reduce organic matter or terminal polishing requirements, artificial cell/enzyme carrier technology offers a precise and controllable solution. By encapsulating catalysts such as specific oxidase, peroxidase, or engineered PETase in hydrogels, polymer capsules, or nanocarriers and circulating them in the reactor, experimental and pilot data show that the

removal rate can often reach or exceed 90% under targeted conditions. It is also easy to recover and chemically inactivate, significantly reducing the risk of environmental exposure to living genes.

1.3 System analysis of technical performance evaluation and adaptation scenarios

As a frontier field of synthetic biology applications, bioelectrochemical technology presents unique advantages in distributed processing and energy recovery. The latest mobile microbial fuel cells have recorded peak power densities of more than 10 mW cm^{-2} and current densities of tens of mA cm^{-2} under experimental conditions, showing that they can simultaneously purify organic matter and recover electrical energy in distributed treatment or remote power supply scenarios, providing the possibility of achieving energy neutrality and even positive energy benefits of wastewater treatment.

Comprehensive comparative analysis shows that the significant advantages of synthetic biology technology are reflected in its high targeting, low energy consumption, and resource utilization path, as well as modular and rapid iteration engineering capabilities. By designing metabolic pathways or specific enzymes, deeper removal of trace amounts of difficult-to-degrade substances can be achieved, solving the problem of insufficient selectivity of traditional technologies in the treatment of low-concentration complex pollutants. By combining biocatalytic modules with ecosystem engineering, the synergistic effect of nutrient recovery and energy recovery can be realized while ensuring treatment efficiency, providing technical support for the circular economy model. Encapsulating catalytic functions as recyclable artificial units provides a realistic path to risk-controlled practical deployment, minimizing environmental risks while maintaining technical effectiveness.

Different synthetic biology technologies have clear adaptation scenarios and application boundaries. Living Machine is suitable for small and medium-sized places with high landscape requirements; Engineered bacterial technology is more suitable for the targeted treatment of specific pollutants in industrial wastewater. Microalgae systems have advantages in nutrient recovery and biomass production; enzyme carrier technology plays a unique role in the terminal treatment of high-value trace pollutants; Bioelectrochemical technology is suitable for distributed processing and energy self-sufficiency applications.

2. Biosecurity risks and prevention and control paths of synthetic biological sewage treatment

As an emerging interdisciplinary field, synthetic biology has shown great application potential in wastewater treatment in recent years. By designing and constructing new biological systems, synthetic biology provides innovative solutions for wastewater treatment, which can efficiently degrade harmful substances such as organic pollutants and heavy metals, achieve sewage purification, and promote resource recycling. However, with the rapid development of technology and the gradual advancement of practical applications, its potential biosecurity risks have also attracted increasing attention. How to effectively manage security risks while giving full play to technological advantages has become a core issue in the sustainable development of this field.

2.1 The dual characteristics of technology application progress and risk challenges

The application of synthetic biology in wastewater treatment is moving from laboratory research to practical application, showing significant technological progress. Scientists have modified microorganisms to have specific metabolic pathways, which can specifically target the degradation and transformation of specific pollutants in sewage. For instance, the UK's Centre for Environmental Biotechnology Innovation (EBIC) is undertaking a five-year project to establish the necessary testing and biosafety frameworks to deploy pollutant-targeting microorganisms in the UK's water and wastewater systems. The project is funded by the UK Research and Innovation Biotechnology and Biosciences Research Council and involves several leading UK universities, each working on a different aspect of microbial editing, management, and practical application. In addition, Spanish energy producer Repsol is applying synthetic biology techniques to design and develop a microorganism for the production of advanced biofuels based on novel anabolic pathways.

However, technological progress has also brought unprecedented risks and challenges. These modified microorganisms carry synthetic genes, and there are still many uncertainties about their behavior patterns and long-term effects in complex environments. Unlike traditional physicochemical treatment techniques, synthetic biology involves the release and application of living microorganisms, and its potential risks have the special properties of self-replication, diffusion, and dissemination, and ecological impact.

2.2 The complexity of identifying and assessing major biosecurity risks

The biosecurity risks faced by synthetic biological wastewater treatment are mainly concentrated in three levels, each with complex impact mechanisms and assessment challenges.

The risk of gene escape and horizontal gene transfer is the most direct safety hazard. Modified microorganisms may escape to the natural environment during sewage treatment, and the synthetic genes carried by these microorganisms may be passed on to other microorganisms through horizontal gene transfer, leading to the spread of genes in microbial populations. This gene spread can alter the structure and function of microbial communities, with unpredictable effects on ecosystems, especially when these genes confer new metabolic abilities or resistance traits on microorganisms.

The risk of niche intrusion and ecological imbalance focuses on the potential impact of modified microorganisms on natural ecosystems. Synthetically biologically modified microorganisms may occupy specific ecological niches in wastewater treatment environments and have the potential to continue to survive and reproduce in their natural environment after release. They may compete with microorganisms in natural ecosystems for resources and may even become invasive species, disrupting the original ecological balance. This ecological risk assessment is particularly difficult because the complexity of ecosystems and the diversity of microbial interactions make accurate predictions challenging.

More complex are the systemic challenges of biosecurity risk assessment itself. The application of synthetic biology in wastewater treatment involves complex biological systems and environmental interactions, and the behavior and influence of microorganisms are affected by various factors, such as environmental conditions, interactions with other organisms, etc. This multi-factor interaction faces limitations in traditional risk assessment methods, necessitating the development of new assessment frameworks and methodologies to address these challenges.

2.3 Build a comprehensive biosecurity prevention and control system

In the face of complex risks and challenges of synthetic biological wastewater treatment, it is necessary to build a comprehensive prevention and control system covering the regulatory system, technical prevention and control, and social governance, and form a multi-level and multi-dimensional safety and security network.

Strict supervision and an approval mechanism is the institutional foundation of the prevention and control system. It is necessary to establish a special regulatory body responsible for approving and supervising relevant projects, which should have professional technical knowledge and capabilities to conduct a comprehensive risk assessment of synthetic biology-modified microorganisms. A sound approval process should include a review of the detailed description, intended use, potential risk assessment, and risk control measures of the modified microorganism. During the approval process, project applicants should be required to provide sufficient experimental data and risk assessment reports to prove that the application of modified microorganisms in sewage treatment is safe and will not cause harm to the environment and human health.

Biosecurity design and technical prevention and control are the core means of risk control. In the process of synthetic biology modification of microorganisms, biosafety design principles should be followed, such as designing microorganisms to survive and function under specific environmental conditions, so as to effectively prevent microorganisms from surviving and spreading in non-target environments. At the same time, modern biotechnology means, such as CRISPR-Cas9 and other gene editing technologies, should be made full of to further optimize and prevent and control modified microorganisms, so that they can be more stable and safe in the process of sewage treatment. These technical control measures should be built into the microbial design to form multiple safety mechanisms.

Public participation and social governance are indispensable and important components of the prevention and control system. A public participation mechanism should be established to allow the public to participate in the decision-making process of synthetic biology in sewage treatment, and to listen to the opinions and suggestions of the public by organizing public hearings and public consultations, so as to make decision-making more scientific and democratic. At the same time, it is necessary to strengthen the public's understanding and understanding of synthetic biology and its application in sewage treatment, and popularize the basic knowledge, potential risks, and prevention and control measures of synthetic biology to the public through popular science activities and lectures, so as to cultivate a rational social cognitive environment.

In general, synthetic biology has broad application prospects in sewage treatment, but the complexity and particularity of its biosecurity risks require us to establish a comprehensive prevention and control system that matches it. Through the organic combination of institutional norms, technical prevention and control, and social governance, the biosecurity risks of synthetic

biology in sewage treatment can be effectively prevented and controlled, and the coordinated development of sewage treatment and ecological environmental protection can be realized while ensuring the vitality of technological innovation, laying a solid foundation for the responsible development of technology and maximizing social benefits.

3. Stakeholder rights and responsibilities coordination mechanism for synthetic biological sewage treatment

The research and development, and application of synthetic biology sewage treatment technology is a complex social project involving the interests of scientific research, industry, government, and the public. The technical challenges, security risks, and social impacts it faces far exceed the ability of a single entity to respond. Therefore, building a scientific and effective power and responsibility coordination mechanism and promoting in-depth communication and close cooperation among all parties is the cornerstone of the system to ensure the healthy development of technology and serve the public interest.

3.1 Major stakeholders and their core rights and responsibilities

The roles of various stakeholders in the whole technology chain are differentiated, and it is necessary to clarify their rights and responsibilities to achieve unification. As the source of technology, scientific research institutions and scientists enjoy academic freedom, the right to use scientific research funds and innovate achievements, and at the same time assume the primary ethical and safety responsibilities: they need to strictly abide by laboratory biosafety norms, carry out forward-looking risk assessment (such as the risk of engineering bacteria leakage) in the early stage of research and development, develop endogenous biosafety technologies simultaneously, and objectively convey technical details to regulatory agencies, interpret values and risks to the public, and avoid information asymmetry.

Technology development enterprises and industries are the core carriers of technology transformation, enjoying intellectual property rights, technical benefits and industrial development rights, and at the same time need to bear the responsibility of full-process security: in the product development stage, multi-scenario safety and effectiveness tests need to be completed to form

traceable data; After listing, establish a full life cycle traceability and emergency response mechanism, quickly deal with problems, and carry out technical and safety training for end users to ensure standardized application.

The core responsibility of government and regulatory agencies such as environmental protection and science and technology is to balance "innovation and prevention and control": it is necessary to build a flexible and dynamic regulatory framework to avoid excessive inhibition of innovation or regulatory lag, and at the same time strengthen professional capabilities (such as biosafety assessment) to ensure transparent approval, unified standards, and fair law enforcement, to provide a stable policy environment for technological development.

As technology applications, end users such as sewage treatment plants and pollutant discharge enterprises have the right to obtain safe, efficient, and reasonable technology and services, and need to fulfill operational responsibilities in the application: use products according to specifications (such as controlling the dosage of bacterial agents) to ensure the normal operation of physical isolation facilities (such as reactor sealing devices); Cooperate with supervision and inspection, report equipment failures, abnormal bacterial agents, etc. promptly, and assist in disposal to reduce risks.

In addition, the public has the right to a clean environment and has the right to know, participate and supervise the application of technology, while environmental non-governmental organizations (NGOs) play an independent supervisory role: the public needs to rationally understand the technology, participate in public discussions and hearings, and express concerns in accordance with the law (such as water safety); Carry out third-party environmental assessments, convey information and feedback to all parties, and promote technology to meet the public interest.

3.2 The path to build a coordination mechanism for rights and responsibilities

In order to achieve coordination among all parties, it is necessary to build a mechanism from four dimensions to form a pattern of "multi-party participation and risk sharing". The establishment of a normalized multi-party dialogue platform is crucial, and it is necessary to regularly organize forums or roundtables for scientists, entrepreneurs, policymakers, NGOs, and the public to discuss key issues such as technology direction, risk standards, and regulatory policies. In response to controversial issues such as the ecological risks of engineering bacteria, temporary consultations can be initiated to build consensus and provide public opinion support for decision-making.

In addition to core trade secrets such as undisclosed patents, the disclosure of technical safety assessment reports, environmental monitoring data such as microbial changes and degradation

efficiency of water bodies, and application cases such as processing scale and cost feedback should be disclosed. Clarify the responsibilities and update frequency of information disclosure of enterprises, governments, and scientific research institutions, ensure that all parties can inquire and supervise, and avoid a crisis of trust.

Participatory governance throughout the life cycle needs to embed stakeholder participation into the whole technology process, introduce ethical and social impact assessment in the R&D project establishment stage, and invite the public to participate in the design of indicators. hold hearings during the pilot test and approval stages to listen to user needs and public concerns; Establish a feedback mechanism in the later stage of application to promote technology iteration and policy optimization to achieve dynamic governance.

The legal responsibility and risk sharing mechanism needs to refine the liability clauses in the technical field under the existing legal framework, and clarify the legal liability and compensation standards of each entity in accidents such as ecological damage caused by bacterial agent leakage. At the same time, we will explore the tripartite risk compensation mechanism of "government + enterprise + insurance company", set up an ecological compensation fund, and implement safety liability insurance to ensure that the environment can be quickly repaired in the event of an accident, compensate victims, and ensure the continuity of technological development.

Therefore, the value realization of synthetic biological wastewater treatment technology requires both technological breakthroughs and the maturity of the treatment system. By building a mechanism with clear rights and responsibilities, collaboration and efficiency, it can not only provide a relaxed environment for innovation, but also ensure that technology is in line with the public interest, and is expected to ultimately achieve a win-win situation for technological progress, environmental improvement and social well-being.

Initiative on Ethical Guidelines and Risk Prevention for the Application of Synthetic Biology in Marine Environmental Governance

Written by XJTLU-China

1. Introduction

The ocean covers approximately 71% of the Earth's surface and is a crucial support system for the global carbon cycle, climate regulation, and biodiversity. However, global marine ecosystems are facing unprecedented pressures: approximately 8 million tons of plastic waste flow into the ocean annually, causing marine organisms to ingest it and die¹. The frequency of red tides and green tides caused by eutrophication is continuously increasing, leading to a 10%–15% decline in coastal fishery production, severely affecting fishermen's livelihoods². Ocean acidification is causing coral reef degradation and slowed growth of shellfish, threatening the entire food chain.

Synthetic biology offers potential solutions through the design of engineered bacteria or genetically modified organisms. For example, using genetically modified microorganisms to degrade plastics, control harmful algal blooms, or monitor seawater pollutants. However, introducing novel species into open environments may bring about ecological, ethical, and social risks, necessitating systematic consideration at scientific, institutional, and societal levels.

2. Potential risks and ethical issues of introducing novel species

2.1 Ecological imbalance and species invasion

Engineered bacteria may compete with natural microorganisms for resources, leading to an imbalance in the original microbial community. For instance, if engineered bacteria rapidly consume specific nutrients, it could cause a sharp decline in phytoplankton populations, subsequently affecting fish and other higher trophic level organisms³.

According to international biodiversity conservation reports, approximately 12% of invasive species cause a decline in local ecosystem function or species extinction annually⁴. Although engineered bacteria differ from natural invasive species, their potential diffusion patterns may be similar.

2.2 Horizontal gene transfer

Engineered bacteria may transfer exogenous genes to natural bacterial populations through plasmids, phages, or other horizontal gene transfer mechanisms. Experiments have shown that certain antibiotic resistance genes can spread throughout a community in marine microorganisms within weeks to months⁵.

Horizontal gene transfer could not only alter microbial ecological functions but also affect biochemical cycles, such as the efficiency of the nitrogen and carbon cycles⁶.

2.3 Uncontrolled diffusion and regulatory gaps

Once engineered organisms enter open waters, even with designed suicide switches or nutrient dependency constraints, there is a risk of mutational escape⁷. Simultaneously, in transnational waters, due to inconsistent regulatory systems, "regulatory blind spots" may arise, making it difficult to respond promptly to ecological anomalies.

2.4 Ethical dilemmas

2.4.1 Intergenerational impacts

Released engineered organisms may alter the structure of marine ecosystems, disrupt the original balance of the marine ecological environment, and cause irreversible impacts on future generations.

a) Environmental Justice: Coastal communities rely on fisheries for their livelihoods. Ecological imbalances could lead to economic and social losses, while the benefits might primarily be gained by research institutions or companies.

b) Moral Controversy: The public and the scientific community need to consider whether humans have the right to arbitrarily modify and release new species into the natural ocean.

3. Impacts on stakeholders

3.1 Scientific research community

Research teams are both drivers of technological innovation and direct managers of ecological risks. If risk assessment is inadequate, research teams may face legal liability and a decline in public trust. More resources need to be invested in genetic design safety testing, contained experiments, and long-term ecological monitoring, increasing R&D costs.

3.2 Coastal communities and fishing industry practitioners

The ecological risks of modified microorganisms or engineered bacteria could directly impact the marine food chain, potentially reducing fishermen's catch and income by 5%–20%. Conversely, if projects successfully manage red tides or pollution, these groups could benefit from cleaner seawater and higher fishery yields, demonstrating clear social benefits.

3.3 Policymakers and regulatory agencies

There is a need to strike a balance between technological innovation and ecological safety by establishing transnational laws and ethical standards. Simultaneously, they may face pressure from various sides, including research, the public, and environmental organizations, requiring a transparent decision-making process.

3.4 Public and social groups

Lack of transparency or media exaggeration may cause panic, for instance, the notion that "genetically modified organisms will become completely uncontrollable." Transparent risk communication and public participation can increase social trust while promoting positive interaction between technology and policy.

4. Pathways to Solutions

4.1 Scientific-level prevention and control measures

4.1.1 Genetic constraints

a) Use "suicide switch" designs (such as Deadman or Passcode systems), where the engineered organism automatically dies when environmental conditions do not meet preset parameters⁷. Experiments show that such genetic switches can reduce the survival rate of organisms in non-experimental environments to below 0.1%.

b) Nutrient dependency: Design engineered organisms to depend on specific nutrients found only in artificial culture media, ensuring they cannot survive outside the experimental environment. Research indicates that such strategies can completely inactivate bacteria within 48 hours in simulated marine environments.

c) Light-sensitive circuits: Certain designs cause engineered bacteria to undergo suicide under natural light, ensuring they cannot spread long-term in open waters. Experimental data show that light-sensitive suicide circuits can inactivate 99% of bacterial cells within 12 hours.

4.1.2 Physical containment

- a) Closed bioreactors: Operate engineered microorganisms within reactors, preventing their direct release into the marine water body.
- b) Degradable containers: Design containers that degrade naturally within a specific timeframe, where the internal environment supports engineered bacterial growth but ceases to do so upon release.
- c) This dual containment (physical + biological) can reduce the risk of engineered bacterial escape by at least 3 orders of magnitude.

4.1.3 Tiered risk assessment

- a) Laboratory stage: Test microbial growth and suicide switch stability in BSL-1 or BSL-2 laboratories.
- b) Microcosm simulation stage: Simulate strain behavior in miniature marine ecosystem models, monitoring changes in microbial communities, impacts on nutrient cycling, and the potential for horizontal gene transfer.
- c) Pilot stage in local waters: Conduct limited experiments in controlled small sea areas or net pen environments, accompanied by long-term monitoring to ensure ecological impacts are manageable.

4.2 Institutional and ethical level measures

4.2.1 Preemptive ethical review

- a) At the project initiation stage, establish an ethics committee comprising ecologists, sociologists, legal experts, and public representatives to jointly evaluate the project.
- b) Review content should include: ecological risk assessment, identification of conflicts of interest, potential social impacts, and safeguarding the interests of coastal communities.
- c) Example: Some international iGEM teams, when testing microbial plastic degradation projects, first solicited opinions from coastal communities and established feedback mechanisms.

4.2.2 International cooperative regulation

- a) Establish transnational regulatory standards clarifying the permitting process for applying synthetic biology in open waters.
- b) For projects involving cross-border waters, recommend adopting joint regulatory mechanisms, such as approval by an international committee under the UN Ocean Science framework.
- c) Implement international supervision over experimental design, risk control, emergency response plans, and other aspects to achieve transparent management.

4.2.3 Public participation and transparent communication

a) Use methods such as science popularization lectures, public hearings, and online information releases to help stakeholders understand project goals, risks, and prevention measures.

b) Establish public feedback and complaint mechanisms, allowing community residents to participate in the decision-making process.

c) Case study: In some European coastal areas, before conducting experiments with synthetic microorganisms for algal management, ecological risk assessment reports were released in advance, and public consultations were accepted, significantly increasing public trust.

5. Our initiative

We propose that when utilizing synthetic biology approaches to address marine environmental issues, ecological safety must always be the bottom line, maintaining a "zero-tolerance" attitude towards any risk that may cause environmental damage. On this basis, dual protection mechanisms must be adopted, implementing control measures at both the genetic design and physical containment levels to reduce the possibility of engineered organisms becoming uncontrollable from the source and physical conditions. Simultaneously, we emphasize open and transparent scientific communication; research teams should proactively address public concerns to avoid misunderstandings and panic caused by information asymmetry; and through continuous science popularization and information disclosure, enhance social trust. We also call for promoting international cooperation and joint ethical development, establishing unified global ethical and regulatory standards for marine applications of synthetic biology to prevent loopholes and risk spillover due to regulatory differences. Finally, we believe it is essential to focus on vulnerable groups, especially coastal communities reliant on fisheries and marine resources for survival, fully considering their interests in project design and application to ensure the realization of environmental justice and social equity.

Ethical Considerations of Different Water Pollution Monitoring Technologies

Written by ZJU-China

1. Introduction

Water pollution monitoring is a vital component of ensuring environmental safety, public health, and sustainable development. With advances in molecular biology, artificial intelligence, and remote sensing, water quality monitoring methods have become increasingly diversified. However, while applying these technologies, attention must not only be given to scientific accuracy and efficiency, but also to the ethical implications behind them, spanning environmental ethics, public health, social equity, and data protection.

2. Mainstream monitoring technologies

In recent years, water pollution monitoring technologies have developed in a diversified manner, with different methods offering distinct advantages in sensitivity, coverage, and application scenarios. Traditional physicochemical monitoring methods—such as pH, conductivity, dissolved oxygen (DO), chemical oxygen demand (COD), and heavy metal ion detection—have long been widely applied in environmental regulatory systems due to their reliability and maturity. According to reports from the United Nations Environment Programme, more than 70% of global water quality monitoring data still relies on such approaches.

In contrast, the development of molecular biology and biosensing technologies provides new opportunities for detecting trace pollutants and biological risks. Environmental DNA (eDNA) detection has recently been used to track invasive species and microbial contamination in water sources. For example, in the cyanobacteria bloom event at Lake Taihu, researchers used eDNA methods to analyze algal community structures, finding that the sensitivity was 3–5 times higher than that of traditional microscopic observation.

Remote sensing and geographic information systems (GIS) enable large-scale water monitoring

capabilities. NASA's MODIS satellite has long been applied to monitor eutrophication and algal blooms in lakes and coastal waters. In the 2011 large-scale algal bloom in Lake Erie, remote sensing successfully achieved real-time tracking of a bloom covering over 3,000 km². However, limitations in resolution and water optical properties mean that such methods remain insufficient for detecting trace pollutants and groundwater contamination.

Furthermore, with the rapid advancement of artificial intelligence and big data, machine learning-based pollution prediction and intelligent sensor networks have become emerging research hotspots. Studies have shown that using deep learning models trained on water quality data from the Yangtze River basin in China can reduce the root mean square error of major pollutant concentration predictions by over 25%. Meanwhile, smart sensor networks have been deployed in smart city water management systems. For example, Singapore's PUB water agency has installed over 200 online sensing nodes, enabling real-time monitoring and automatic alerts for organic pollutants in the water supply system. However, these methods are highly dependent on data quality and algorithmic transparency. Without robust regulatory mechanisms, they may raise risks of data bias and accountability concerns.

3. Ethical considerations

From an environmental ethics perspective, water pollution monitoring aims to protect ecosystems, yet the application of monitoring technologies may inadvertently introduce new risks. Traditional physicochemical monitoring often requires large-scale water sampling, which may disturb aquatic ecosystems. For example, a wetland ecological survey in California found that excessive sampling caused measurable disturbances to zooplankton community structures. Additionally, with the advancement of molecular biology and biosensor technologies, some studies have explored the use of genetically modified microorganisms as in situ monitoring tools. While such approaches improve sensitivity, they pose risks of gene leakage, species competition, and even "secondary pollution." Simulations of genetically engineered *E. coli* in lake environments revealed that they could survive for more than 14 days, raising ecological safety concerns. Thus, a key ethical question is whether monitoring methods might introduce new environmental threats while aiming to protect ecosystems.

From a public health ethics perspective, water pollution monitoring results directly affect community health, making the transparency and timing of information disclosure critical. If serious pollution is detected but disclosure is delayed, residents' drinking water safety may be endangered; if information is released too hastily, public panic may ensue. A typical case is the Flint water crisis in Michigan, USA, where government and institutions delayed disclosing lead contamination results. As a result, over 100,000 residents consumed contaminated tap water for an extended period, leading to significantly elevated blood lead levels among children. This incident highlights the profound impact of transparency on public risk perception and trust, as well as the ethical dilemma of balancing "avoiding panic" with "ensuring the right to know."

From the perspective of social equity and justice, advanced monitoring technologies are often costly, potentially exacerbating disparities in water management capacity between regions. Reports from the World Health Organization indicate that in low-income countries, more than 60% of water monitoring projects rely on international aid, compared to less than 10% in developed countries. Such imbalances not only affect access to environmental data but may also exacerbate environmental injustice in policymaking and resource allocation. For example, in some South Asian cities, insufficient monitoring capacity has left residents exposed to untreated industrial wastewater for years, while policy interventions lagged. Ensuring fair access to environmental data and the equitable application of monitoring outcomes has thus become a pressing issue of social justice.

In terms of data ethics and privacy protection, the growing use of remote sensing and big data in water monitoring raises concerns about data security. Remote sensing may involve pinpointing industrial discharge sites, and unauthorized disclosure could harm business interests or trigger legal disputes. Similarly, real-time data collected by AI and sensor networks may include geographic locations, sampling sites, or even household water usage, which—if mismanaged—could be abused. The European Union's General Data Protection Regulation also applies to environmental data, emphasizing that data collection and sharing must be transparent, lawful, and minimized to prevent violations of personal or organizational privacy.

Finally, regarding technology governance and accountability, the increasing role of AI and smart sensors in water quality monitoring makes prediction errors and false alarms inevitable, raising disputes over responsibility. For instance, in one Chinese city, an intelligent sensor network mistakenly interpreted rain-induced turbidity as heavy metal contamination due to improper algorithm parameters. This misjudgment led local authorities to launch emergency pollution

responses, causing resource waste and public alarm. This case demonstrates that without clear accountability frameworks and multi-stakeholder governance mechanisms, technological risks are difficult to manage. Therefore, it is essential to establish a shared responsibility system involving technology developers, users, governments, and the public to ensure the reliability and sustainability of monitoring technologies.

4. Conclusion

While different water pollution monitoring technologies improve efficiency and accuracy, they also bring complex ethical challenges. Scientists, policymakers, and the public should work together to address environmental ethics, health rights, social equity, and data protection, seeking a balance between scientific progress and ethical responsibility. Only by doing so can monitoring technologies truly serve the goal of harmonious coexistence between humanity and nature.

Ethics of Synthetic Biology: Prediction, Intervention, and Responsibility in HABs Management

Written by SUSTech-OCEAN

1. Ethical dilemmas of prediction

With the rapid advancement of science and technology, synthetic biology, as an emerging interdisciplinary field, is providing new ideas and methods for solving global environmental problems. Harmful Algal Blooms (HABs), as a severe marine ecological disaster, have a tremendous impact on marine ecosystems, fishery economies, and human health. In recent years, the application of synthetic biology technologies in HAB prediction has gradually garnered attention. By designing and constructing new biological systems, it offers novel technical means for HAB forecasting. For example, gene editing technology can be used to develop biosensors that sense and respond to HAB-causing algae. Recently, the Key Laboratory of Experimental Marine Biology at the Chinese Academy of Sciences utilized CRISPR gene editing technology to reveal a calcium-mediated intracellular signaling pathway in marine diatoms responding to population density changes, enhancing our understanding of population dynamics during diatom bloom outbreaks and thereby providing potential for early warning of abnormal algal growth^[1]. Furthermore, synthetic biology can determine the species, distribution, and genetic characteristics of HAB algae by analyzing their gene sequences. This method possesses high accuracy and reliability, offering an important basis for the monitoring and early warning of HAB algae^[2]. However, the more technically feasible this powerful predictive capability becomes, the more prominent its social and ethical challenges are. It compels us to consider: Who has the right to control and distribute this predictive power? What risks are hidden behind it? This chapter aims to dissect the profound contradictions arising from this technological application at the levels of data ethics, social equity, and public perception.

1.1 Ethical risks arising from data and algorithms

Although synthetic biology holds significant application value in HAB prediction, it also entails

potential risks. The realization of predictive capability is highly dependent on data and algorithms, which gives rise to a new ethical triad: ownership, privacy, and boundaries. Who owns the data collected by biosensors deployed in public waters? Is it the technology company that developed them, the government agency that deployed them, or does it belong to all humanity? If a private company possesses the most accurate HAB prediction data and charges users for access, issues concerning the commodification and attribution of rights emerge. Relevant reports indicate that such unjust distribution of "predictive power" has already occurred, where aquaculture farms in affluent areas can obtain warnings, while small-scale fishermen are excluded^[3].

Continuous biological monitoring can indeed be viewed as a form of intrusive surveillance of the natural environment. Although the purpose is beneficial, such as predicting HABs, this practice essentially establishes an uninterrupted data collection system within the natural environment. The purposes for which this environmental data is used may likely extend beyond environmental protection. There is a risk it could be utilized for military or commercial espionage. For instance, plankton communities are highly sensitive to changes in the underwater environment; minor disturbances, noise, or emissions produced by submarine navigation can alter plankton distribution or behavior. By analyzing such data, it is theoretically possible to infer submarine activity in a given area, which is clearly a military application^[4,5]. The types of data involved in synthetic biology are rich and diverse, with different data types having varying privacy protection needs and risk profiles. The complexity of privacy protection is also a significant consideration.

1.2 Impact of synthetic biology on different social groups

The application of synthetic biology for HAB prediction affects different social groups, each with its own stance and concerns. Balancing these diverse perspectives is key to the rational application of this technology. For scientists, their primary concern is technological breakthrough and innovation. They strive to research more accurate prediction methods, hoping to prove the feasibility of the technology. However, sometimes they may become overly focused on the technology itself, neglecting public acceptance or underestimating potential risks. Scientists have a responsibility to remain open and honest, not only introducing the benefits of the technology but also clearly explaining potential uncertainties and risks, and proactively conducting comprehensive environmental assessments. For fishermen and aquaculturists, they are the ones directly affected by

the technology. They rely on early warning information to protect their livelihoods, so they value accuracy above all. If the system issues a false alarm, they suffer losses from unnecessary work stoppages; if it fails to warn (a missed alarm), they may face significant economic losses. Therefore, the reliability of the technology and the integrity of compensation mechanisms are crucial for its application in real-world scenarios.

1.3 The ‘Natural’ vs. ‘Artificial’ debate in public perception

Beyond specific social groups, societal cognition is also impacted. The widespread societal notion that "natural is superior to artificial" leads many to instinctively view the release of engineered microorganisms as "pollution" of nature. This mindset often causes the public to overestimate the "risks of action" associated with technological application (e.g., engineered bacteria running out of control), while underestimating the "risks of inaction" posed by HABs themselves^[6], creating a cognitive asymmetry that hinders the rational application of technology.

2. Hazards of biological methods: ethical considerations and potential impacts

Harmful Algal Blooms (HABs), as a severe marine ecological disaster, have profound negative impacts on marine ecosystems, fishery resources, and the lives of coastal residents. In recent years, the development of synthetic biology has provided new ideas and methods for HAB prevention and control. However, the application of these biological methods also raises numerous ethical issues and potential risks, requiring in-depth discussion and careful evaluation. This section will focus on the hazards of biological methods using synthetic biology for HAB control, including ethical considerations of introducing new species into nature, ethical issues arising from the failure of kill switches, and other potential impacts.

2.1 Ethical considerations of introducing new species into nature: natural balance and biodiversity

Ecosystems are complex networks of interactions where species exist in a delicate balance. Introducing new biological species through synthetic biology means may disrupt this balance. For

example, genetically modified algae introduced to control HABs might compete with native algae for resources, thereby affecting the survival and reproduction of native species. This could not only alter the species composition of the marine ecosystem but also trigger chain reactions affecting the entire ecosystem's function. From an ethical perspective, we have a responsibility to protect biodiversity in nature and maintain the integrity and stability of ecosystems. Therefore, comprehensive ecological risk assessments must be conducted before introducing new species to ensure no irreversible damage is caused to the natural ecosystem.

2.2 Ethical issues arising from kill switch failure

Kill switches are an important safety mechanism in synthetic biology designed to prevent engineered organisms from surviving and reproducing in the environment after completing their task. However, kill switches are not absolutely reliable; if they fail, it could lead to engineered organisms becoming uncontrollable in the environment. These out-of-control organisms might proliferate extensively in the environment, compete with native organisms for resources, and potentially transfer engineered genes to other organisms through horizontal gene transfer, thereby triggering uncontrollable ecological risks. For instance, if the kill switch fails in engineered algae, it might proliferate massively in the ocean, forming new algal blooms, or even hybridize with other algae, producing new algal species with greater adaptability and harmfulness. Such uncontrollable ecological risks could not only cause severe damage to marine ecosystems but also negatively impact human health and economic interests. From an ethical standpoint, such uncontrollable ecological risks are unacceptable. In HAB control, we must ensure the reliability of kill switches, adopt multiple safety measures, and prevent engineered organisms from becoming uncontrolled in the environment.

2.3 Other potential impacts

2.3.1 Long-term ecological effects

The application of synthetic biology in HAB control may have long-term ecological effects. Although certain control effects might be achieved in the short term, it could have profound impacts on marine ecosystems in the long run. For example, engineered algae might persist in the environment long-term, altering water chemistry and nutrient cycling, thereby affecting the structure

and function of the entire marine ecosystem. These long-term ecological effects might take years or even decades to manifest. Therefore, long-term ecological monitoring and assessment are necessary during HAB control to promptly identify and address potential ecological issues.

2.3.2 Genetic pollution

Organisms modified through synthetic biology might transfer engineered genes to other organisms through mechanisms like horizontal gene transfer, leading to issues of genetic pollution. Such pollution could alter the gene pool and affect evolutionary processes. Ethically, we have a responsibility to protect the natural gene pool and maintain the natural evolutionary processes of organisms. During HAB control, effective measures must be taken to prevent genetic pollution, ensuring natural gene transfer and evolutionary processes remain undisturbed.

3. A framework of responsibility ethics for synthetic biology in HAB control

Releasing engineered organisms into the ocean is not only a technical challenge but also requires a robust ethical framework for regulation to prevent unintended harm and ensure equity. Therefore, this chapter aims to construct a responsible ethics framework addressing three core questions: How to apportion responsibility between historical and current actors? How to build a multi-party governance system beyond the "polluter pays" principle? How to ensure the fair distribution of benefits and risks? Without such a framework, even the best technology could foster new ecological and social injustices.

3.1 Acknowledging historical pollution, regulating current actors

HABs are the negative outcome of long-term accumulated pollution, their responsibility rooted in history. Between 1980 and 2010, industrialization processes along China's coast discharged large amounts of nitrogen and phosphorus pollutants, a major cause of current HABs. Furthermore, phosphorus in sediments can be released continuously for decades, creating an "environmental debt" difficult to repay (Zhou et al., 2019). Due to factors such as lack of early regulation, defunct polluting enterprises, and the transfer of international industrial chains, the original responsible parties are often absent or unclear (Cloete et al., 2015).

Simultaneously, emerging pollution sources and climate change complicate responsibility attribution. Residual feed and waste from aquaculture drastically increase nitrogen loads in local waters (Anderson et al., 2021), while global warming extends the algal growth season, exacerbating the duration and intensity of HABs (IPCC, 2023). These current responsible parties are dispersed, and the responsibility for climate change is global, making it difficult to penalize single actors. Therefore, a just ethical framework must acknowledge these historical roots and cannot place the entire burden on current, visible actors.

3.2 Shifting prevention responsibility: from ‘polluter pays’ to ‘multi-party governance’

The traditional "polluter pays" principle is insufficient to address the complexity of HAB control. A collaborative responsibility network involving government, industry, research institutions, and the public must be constructed.

3.2.1 Government

Governments need to play a core regulatory role but face dual challenges of policy implementation and regulatory gaps. For example, China has reduced land-based pollution through cross-watershed governance, but due to local protectionism during implementation, nitrogen and phosphorus reduction targets were only 60% met (Wang et al., 2020). Meanwhile, globally, specific regulations for the environmental release of engineered bacteria are lacking.

3.2.2 Industry

Corporate responsibility must extend beyond compliance obligations. The aquaculture industry needs to implement ecological compensation, such as Norway's "aquaculture license auction" model (Hansen et al., 2022); biotechnology companies should be ethically constrained to avoid "technological colonialism" after patent commercialization and should be compelled to share technology with low-income communities.

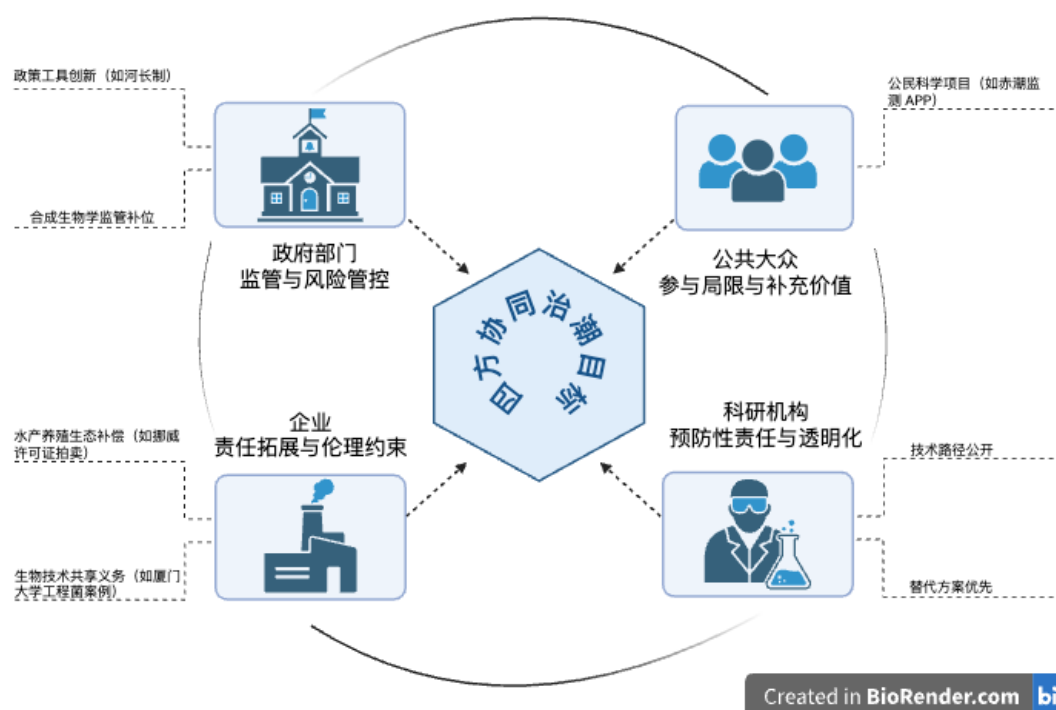
3.2.3 Research Institutions

Research institutions bear a preventive responsibility. They must fulfill obligations of technological transparency by disclosing the design pathways and stability data of engineered bacteria, and adhere to the precautionary principle, prioritizing non-genetic editing alternatives.

3.2.4 Public

While public participation can enhance individual responsibility, there is a need for vigilance against its potential to obscure systemic responsibility—after all, industrial and agricultural pollution far exceed household discharge (CEFAS, 2021).

Successful cases (such as the "Blue Bay" remediation in Ningbo Meishan Bay) have demonstrated that a governance model led by the government, supported by research, and involving multi-party collaboration can effectively prevent and control HABs, achieving a win-win situation for ecology and the economy.



3.3 The environmental justice dilemma of synthetic biology: misalignment of beneficiaries and risk bearers

The application of synthetic biology raises severe environmental justice issues: Are benefits and risks being distributed fairly? Currently, the answer appears negative.

In the short term, tourism and fisheries are direct beneficiaries. However, in the long term, the biggest winners are the biotech companies that patent and commercialize technologies developed

with public funding. Their high licensing fees could become a heavy burden for small-scale farmers (Zhang et al., 2023).

Meanwhile, risks are disproportionately transferred to vulnerable communities. Engineered bacteria could disrupt the base of the food web, threatening the livelihoods of traditional fishermen; the release of modified organisms might also violate the ecological beliefs of indigenous peoples, harming their cultural rights. Internationally, intellectual property barriers may hinder the application of the technology in neighboring countries, contravening the benefit-sharing principle of the Convention on Biological Diversity (CBD). If the "solution" for controlling HABs creates a few winners at the expense of the majority, it fails the basic test of environmental justice.

3.4 Integrating intergenerational and international justice: an operational framework for responsibility ethics

Addressing the above dilemmas, our project has integrated the concept of "Responsible Research and Innovation" (RRI) throughout the entire process of technology development and governance coordination from the outset. Technologically, we not only pursue efficient control but also strictly prevent gene diffusion and non-target effects through biosafety designs like "suicide switches," reducing ecological and ethical risks. In terms of governance mechanisms, we reject technological black boxes, adhere to the iGEM spirit of open source, promote transparent sharing of technical solutions, and avoid patent monopolies and unfair benefit distribution. At the social responsibility level, we promote marine science education, community participation, and international cooperation through the Blue Ocean Plan (BOP), enhancing the systemic capacity of the whole society to respond to HABs.

However, the responsibility system for HAB control cannot be fully covered by the project's unilateral practices alone—issues such as tracing responsibility for historical pollution legacies, implementing tiered management and control for synthetic biology technology deployment, and cost and patent distribution in transnational control still require more systematic mechanism design to balance the responsibilities of all parties. Based on this, specific action recommendations can be proposed in three areas: For historical responsibility conversion mechanisms, a "HAB Legacy Fund" could be established, levying special taxes on high-pollution coastal industries (e.g., paper, fertilizer)

from 1980-2010 to fund synthetic biology safety assessment research (referencing the US Superfund model).

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